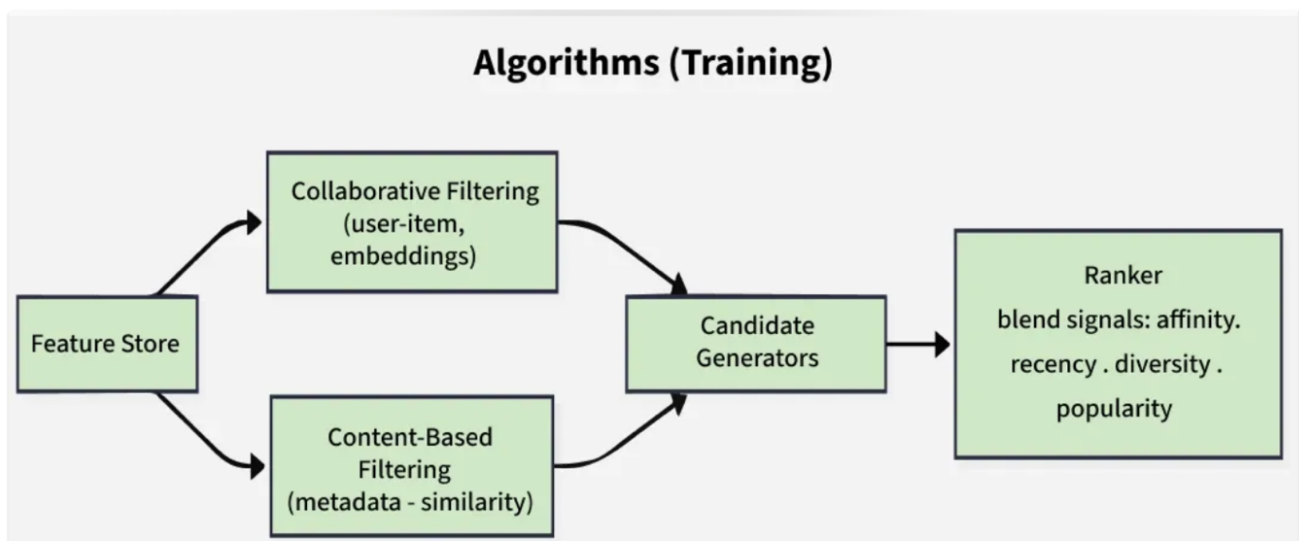
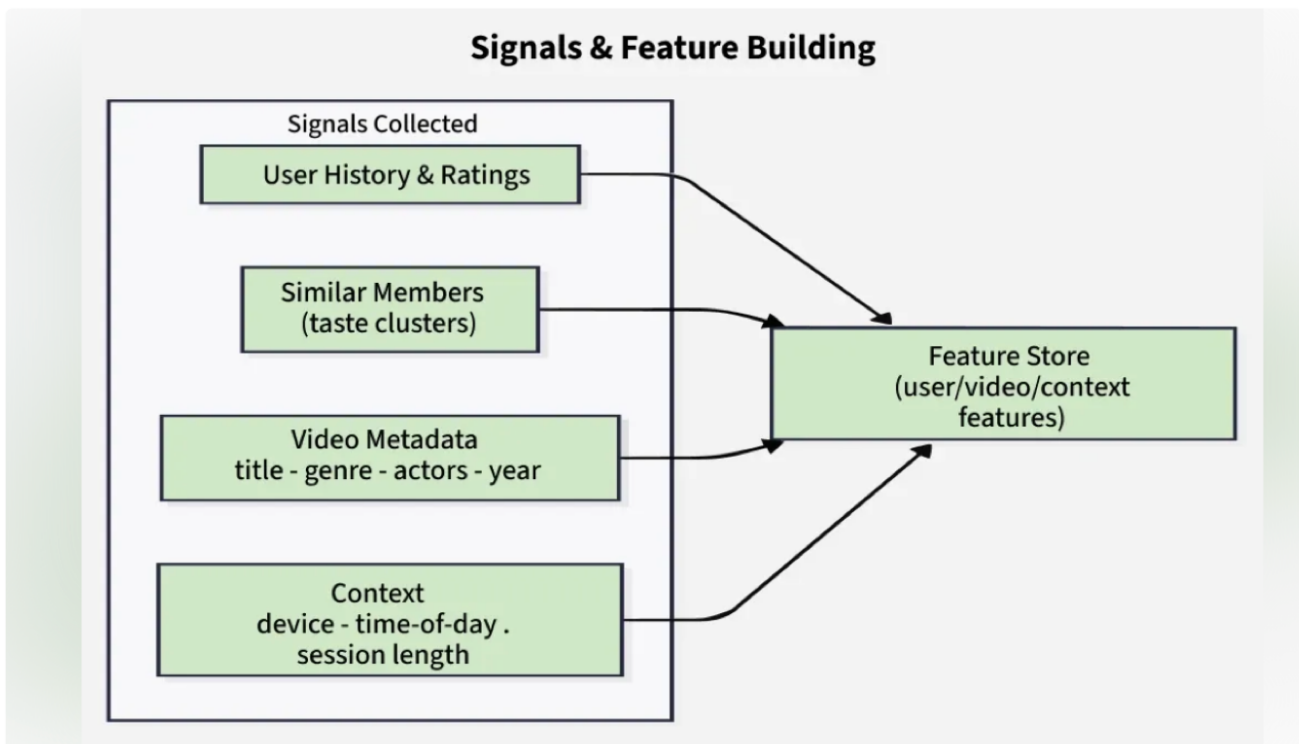


Personalization & Search

Help each profile find something to watch fast (reduce time-to-first-play, increase completion)

Balance relevance, diversity and freshness.



7.1 Signals

- **Behavioral:** plays, pauses, stops, rewinds, completion %, dwell time on title pages, add-to-list, browse depth.
- **Contextual:** device type, time of day, network quality, household profile (Kids, language)

- **Contextual:** device type, time of day, network quality, household profile (kids, language).
- **Item metadata:** genre, cast, director, maturity rating, runtime, audio/subtitle availability, HDR/Dolby tags.
- **Social-like affinity (implicit):** “people who watched X also watched Y.”
- **Quality signals:** prior user satisfaction (finishes), early abandons, rewatch ratio.

7.2 Features & Storage

- **Offline feature store:** heavy aggregates (e.g., long-term genre affinity, recency decay counts) computed in **Spark** and written to a feature store (batch cadence: hours).
- **Online/near-real-time features:** short windows (last session, last 15 min); maintained via stream processors fed from event bus.
- **Join strategy:** at request time, the ranker joins candidate titles with the latest profile features + item features (cached in EVCache).

7.3 Candidate Generation (recall → a few thousand)

- **Collaborative filtering recall:** “viewers like you also watched...” (matrix-factorization or ANN over embeddings).
- **Content-based recall:** same genre/theme/cast embeddings; language/region filters.
- **Business rules:** new & expiring, licensed for region/profile, parental controls.
- **Contextual recall:** device-aware (shorter runtime picks on mobile nights), session-aware (continue watching, episodic next-up).

7.4 Ranking (reduce to a few dozen rows)

- **Two-stage ranker:**
 1. **Lightweight scorer** (GBDT/logistic) trims 2–5k candidates to ~200–400.
 2. **Heavier model** (pointwise/pairwise LTR, neural ranker) orders the final page(s).
- **Objective blend:** predicted play start (pPlay) × expected watch time (EWT) × completion likelihood × diversity boost × freshness.
- **Diversity & saturation controls:** limit near-duplicates (same franchise), spread genres, ensure at least N “exploration” picks.

7.5 Artwork Personalization (why rows look different)

- For the same title, pick artwork variant most likely to earn a click for this profile (cast-centric vs scene-centric poster).
- Model inputs: your past clicks on artwork styles, genre affinity, device (small screens favor high-contrast faces).
- Served inline by the ranker; cached per title/profile for a short TTL.

7.6 Exploration vs Exploitation

- **Multi-armed bandit/epsilon-greedy** on row slots: occasionally test promising but uncertain items.
- **Gradual exploration** where new picks hide near top performers, avoiding high-profile content until...

- **Guardrails:** exploration snare capped; nide poor performers quickly; kids profiles explore only within rating fences.

7.7 Freshness & Latency Budgets

- **Home request budget:** ~100–200 ms server time to assemble a page (excluding network).
- **Cache strategy:** pre-materialize row pages per profile; invalidate on important events (new season drop, strong affinity change).
- **Staleness target:** rows < few minutes old; “continue watching” updated instantly client-side if needed.

7.8 A/B Testing & Feedback Loops

- **Experiment platform:** bucketing at profile level; immutable assignment; ramp & guardrails (QoE, churn risk).
- **Primary metrics:** time-to-first-play, starts/profile/day, average watch time, abandon rate; long-term holdouts to detect over-fitting.
- **Counterfactual checks:** avoid self-reinforcing bias (e.g., only surfacing blockbusters).

7.9 Search (typed queries & voice)

Indexing pipeline

- Tokenize titles, people, genres; normalize accents, synonyms, locales.
- Build inverted index (term → postings with fields: title, cast, synopsis).
- Real-time tier for new/updated titles; merge to warm segments; geo & language fields for filters.

Query path

- Parse & rewrite (spell-correct, synonym/alias expansion, “because you watched X” boosts).
- Retrieve candidates by BM25/ANN; re-rank with behavioral signals (click/play rates, recency, user-author affinity).
- **Filters:** maturity rating, audio/subtitle language, HDR/Atmos availability, runtime buckets.
- **Autocomplete/typeahead:** prefix index + popularity; return entities (titles, people, genres).

7.10 Safety, Policy & Compliance in P13N/Search

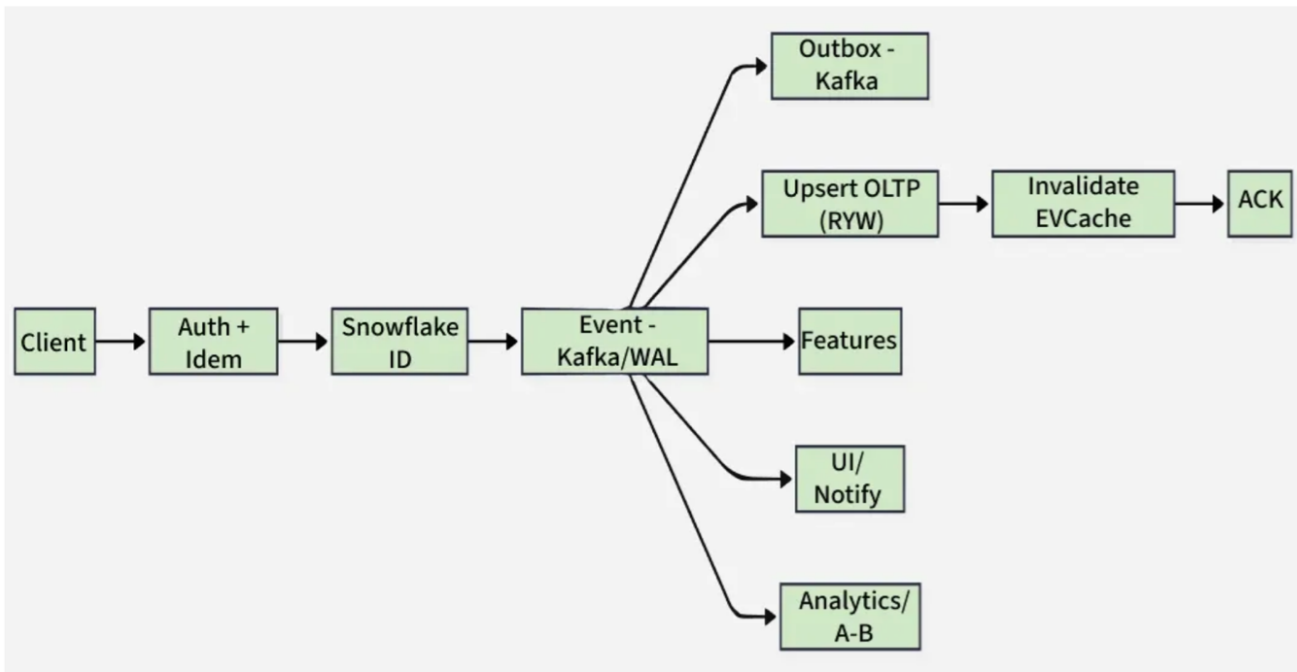
- **Kids profile fences:** strict maturity filters, curated rows; search excludes adult results.
- **Regional licensing:** entitlement checks in both recall and rank.
- **Privacy:** train on **aggregated/anonymous** signals; erase traces on account deletion.

7.11 Failure Modes & Degradations

- **Ranker offline** → fall back to heuristic sort (recency + popularity + user genre mix).
- **Feature store stale** → serve cached row pages; annotate for quick refresh next cycle.
- **Search realtime tier lag** → show latest via a “recently added” side-channel; degrade ranking but keep recall broad

but keep recall broad.

8. Write/Read Paths



8.1 “Add to My List” / Rating (Write)

- **Validate & authorize:** check profile, entitlement, parental controls; idempotency key to avoid duplicates on retries.
- **ID allocation & logging:** allocate Snowflake ID; write append-only event (e.g., `ListItemAdded`, `RatingSet`) to Kafka/WAL first (source of truth).
- **Transactional outbox:** if using OLTP, use outbox → Kafka to avoid dual-write races.
- **State update:** upsert OLTP row (profile list / rating table) with read-your-writes guarantee for the caller (session stickiness or client merge).
- **Cache maintenance:** targeted EVCache invalidation (per-profile list keys, per-title aggregates).
- **Async fans:**
 - **Features:** update user/title features for ranking (e.g., affinity boosts).
 - **Notifications/UI:** nudge UI to refresh “My List”, badges, and row ordering.
 - **Analytics:** counters and A/B beacons (latency-insensitive).

8.2 Home Rows (Read)

- **Hot path:** read **materialized per-profile pages** from EVCache (TTL minutes; **staggered jitter** to avoid thundering herds).
- **Miss path:**
 1. Fetch **candidates** (recent/popular/continue-watching) from feature store + catalog.
 2. **Eligibility filters:** region/DRM availability, maturity rating, device capabilities (codec/HDR).
 3. **Rank:** blend recency, similarity, affinity, and exploration diversity; cap per-brand/show to avoid flooding.
 4. **Assemble page:** join artwork, language tracks, watch-state; **store** in cache; **return** results.
- **Pagination:** opaque cursor (last score + tie-breakers) for stable paging; tolerate **late arrivals**

- **Pagination:** opaque cursor (last score + tie-breakers) for stable paging, tolerate late arrivals (eventual consistency).
- **Freshness:** background refresh on new plays/ratings; **soft real-time** SLAs (seconds) are acceptable.
- **Degradation:** on feature/index lag, fall back to **heuristic rows** (e.g., “Top 10”, “Recently Added”) to preserve UX.

9. Storage Model (Pragmatic)

1. OLTP (RDBMS): identity, billing, entitlements, device registrations, household profiles.

- **Patterns:** strong constraints (uniques, FKs where sensible), multi-AZ, read replicas, PITR/backups.
- **Writes:** short transactions; outbox for change publish; GDPR erasure via tombstones + purge jobs.

2. Wide-column / NoSQL: activity, playback sessions, recent interactions, counters.

- **Access:** partition by profileId/userId for locality; time-bucket hot logs to smooth load.
- **Counters:** approximate (HLL) or CRDT-style with periodic reconcile for exactness.

3. Search index: titles/people/genres; real-time tier (seconds) + archive tier; lifecycle policies & merges.

4. Object storage: video origins, artwork, subtitles; versioned, lifecycle to colder tiers; hash-keyed for dedupe.

5. Feature store: online (low-latency reads by profile/title) + offline (batch); CDC from Kafka → store.

6. Sharding keys:

- **Personalization:** profileId / userId (read-heavy).
- **Catalog edits:** titleId (write ownership clear).
- **Logs/telemetry:** time + region (operational isolation).

7. Multi-region: active/active for browse; edge-biased reads; clear consistency contracts:

- **Strong:** identity/entitlements/payments.
- **Eventual:** home rows, counts, trends.
- **Read-your-writes:** session stickiness or client-side merge to show new actions instantly.

10. E2E Sequence (Play Press → First Frame)

Intent & bootstrap: Client sends Play to **Gateway (Zuul)** with profile/session; device capabilities (codec, HDR, bandwidth hints) attached.

Playback Service checks

- **AuthZ & entitlement:** region, rating, concurrency limits
- **Title availability:** correct audio/subs, CDN readiness

- **DRM policy:** license type, offline eligibility

Manifest & license

- Return manifest (HLS/DASH) with signed CDN URLs (tokenized).
- Parallel DRM license request; include keys for initial renditions and subtitles.

Edge (OCA) selection

- Pick nearest healthy Open Connect node (latency, load, health).
- Fallback: alternate OCA → regional edge → origin shield on errors.

Initial fetch & startup

- Fetch init segment + first media segment(s); target startup < ~2s.
- Conservative initial bitrate based on past sessions / quick throughput probe.
- Prefetch captions & default audio; warm small buffer (e.g., 6–10s).

ABR steady-state

- Measure throughput, variance, buffer; upshift/downshift rungs smoothly (no oscillation).
- Enforce device limits (resolution, codec, HDR, FPS) and data saver settings.
- Stitch CDN token refresh and key rotation seamlessly.

Telemetry & QoE

- Emit events: startup, bitrate switches, stalls, errors, CDN/edge chosen.
- Pipeline: Client → Kafka → S3/Parquet (batch) and ES/Metrics (near-real-time) for QoE, anomaly alerts, and ML features.

Resilience & fallbacks

- Segment timeout → retry same edge → alternate OCA → lower bitrate → shorten read-ahead.
- License/DRM hiccup → quick retry/backoff; if multi-audio fails, degrade to core track.
- Persistent issues → graceful error with retry option; log with correlation ID.

First frame & beyond

- Present first frame; keep buffer safety (e.g., ≥ 2 segments).
- Continue ABR adjustments; background-fetch next subtitles/audio; periodically refresh manifest if needed (live/episodic).